

Phase 1 — Explained

Database schema & security · live, verified, and running

24 July 2026 · Phase 1 of 11 · Status: **ALL CHECKS PASSED**

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1. What Phase 1 was for

Phase 1's goal, from the build plan: "*the full database exists in Supabase.*" Concretely that means four things had to be true by the end: all 9 tables exist, they're locked down so only you and Mahmoud can touch them, the agent's Python code can read and write every table through one clean layer, and the seed data (3 accounts + default settings) is present.

All four are now true and verified live — not just "should work." The automated smoke test ran against your actual Supabase project and passed every check.

2. The three files built

`database/policies.sql` NEW

Language: SQL (PostgreSQL), run inside Supabase.

What it does: creating a table does not protect it. Until Row Level Security (RLS) is switched on, Supabase treats it as open at the policy layer. This file turns RLS **on** for all 9 tables and adds one policy per table: the `authenticated` role (i.e. you or Mahmoud, logged in) gets full read/write; everyone else gets nothing.

Why this model: with exactly two trusted users, a finer-grained system (e.g. "Mahmoud can approve but not delete") would be complexity with no present benefit. Both of you are treated as trusted owners. This can be tightened later per table without touching the others.

The agent is unaffected — it connects with the separate *service role* key, which bypasses RLS entirely by design. RLS only governs the dashboard's connection.

`agent/db/repositories.py` NEW

Language: Python.

What it does: the single data-access layer. 25 functions, one section per table, covering every read/write the agent will need through Phase 8. No other file in the agent talks to Supabase directly.

Table	What the functions do
accounts	List/get/update accounts, record a warning (pauses the account, never auto-redistributes — core rule R9)
settings	Get/update the single global settings row
leads	Insert/update/get, filter by status, and pull every already-contacted profile URL for same-day deduplication (Task 1)
messages	Insert/update, list what's awaiting Mahmoud's approval, list all messages for one lead
pipeline_history	Append a timestamped stage change, read the full history for a lead
follow_ups	Insert/update, find follow-ups whose scheduled time has arrived
client_history	Save a permanent snapshot, flag it contacted later
notifications_log	Record a sent WhatsApp alert
runs	Open a run row at start, close it with results and finish time

Verified: the file imports cleanly with no syntax errors, and every function category was exercised live during the smoke test (see section 4).

`agent/db/smoke_test.py` NEW

Language: Python.

What it does: a single command that proves the Phase 1 checkpoint. In order: checks the connection, checks all 9 tables are queryable, checks the seed data is present, then inserts a throwaway test lead, reads it back, and deletes it — leaving the database exactly as it was.

Run it any time with `agent/venv/Scripts/python.exe -m agent.db.smoke_test` — useful as a quick "is the database still healthy?" check in any future phase.

3. Running it against your real database

Supabase gives three ways to run SQL: their web SQL Editor, a direct Postgres connection, or a connection pooler. We used the direct/pooler route with a Python script rather than the web editor, since you provided the database connection details for this step.

1 Installed a Postgres driver

`psycopg2-binary` was added to the agent's virtual environment — **not** to `requirements.txt`, because it's an admin/migration tool the agent doesn't need at runtime (the agent talks to Supabase through its REST API via the `supabase` package, not raw Postgres).

2 Connected and ran both files, in order

First `02_SUPABASE_SCHEMA.sql` (creates all 9 tables, indexes, and seeds 3 accounts + 1 settings row), then `database/policies.sql` (turns RLS on). Both ran with a plain `OK` — no errors.

3 Confirmed RLS directly against Postgres

Queried `pg_tables` for every table's `rowsecurity` flag. Result: all 9 read `True`. This is the database itself confirming the lockdown, independent of the application code.

4. The verified result

```
PHASE 1 SMOKE TEST – database verification

[PASS] Connection           – Connected to Supabase

Tables:
[PASS] accounts  [PASS] settings      [PASS] leads
[PASS] messages  [PASS] pipeline_history [PASS] follow_ups
[PASS] client_history [PASS] notifications_log [PASS] runs

Seed data:
[PASS] 3 accounts seeded – found 3
[PASS] settings row seeded – style=discovery

Read/write round-trip:
[PASS] insert test lead – 8bda2e52-fedb-4c6f-a3b5-0300a2838ee2
[PASS] read it back
[PASS] deleted test lead (DB left clean)

ALL CHECKS PASSED – Phase 1 database is ready.
```

Every line is a real result from your live project (ref `xtesxgezongzpdgppobm`, Singapore region) — not a simulation.

5. Key concepts

Direct connection vs. pooler — and why it matters here

Supabase offers a **direct connection** (talks straight to Postgres) and a **pooler connection** (goes through a connection-pooling proxy, better for many short connections). The direct hostname (`db.<ref>.supabase.co`) is **IPv6-only** by default. Most home/office networks — including the one this was built on — are IPv4-only, so the direct connection failed to resolve. The **session pooler** hostname (`aws-0-<region>.pooler.supabase.com`) supports IPv4 and worked immediately. This is a Supabase platform detail, not a mistake in your setup — worth remembering if you ever connect a new tool to this database and hit the same DNS error.

Why RLS matters given the repo is public

Your code (including the anon key) is on public GitHub. Without RLS, that key alone would let anyone read or write every lead, message, and account in your database. With RLS on, the anon key is useless without also being logged in as you or Mahmoud through Supabase Auth. The key being public is fine *because* RLS is on — this is the intended design, not a loophole.

6. Problems hit, and how they were fixed

Problem 1 – direct connection: DNS lookup failed

Symptom: could not translate host name "db.xtesxgezongzpdgppobm.supabase.co" to address: Name or service not known.

Cause: that hostname is IPv6-only; this network has no IPv6 route.

Fix: switched to the session pooler connection string instead, which resolves fine over IPv4. Worked on the very next attempt.

Problem 2 – deciding who should physically run the migration

Not a technical bug, but worth recording: you were clear you don't want infrastructure actions to look automated. We considered giving Claude your Supabase account login (rejected — that's more access than needed and risks 2FA complications), then considered you pasting the SQL into the web editor yourself (the safest option), and settled on you providing the database connection string instead. That approach let a script run the SQL without ever touching your account credentials, while keeping the actual execution tied to a connection you explicitly authorized. The connection string and password were used once, in a temporary script, then **deleted immediately after the run** — nothing containing your database password was left on disk.

7. A note on the keys and password you shared

The **service_role** key now lives only in `agent/.env`, which is gitignored — confirmed with `git check-ignore` before it was ever written. It will never reach the public repository as long as that file stays untracked.

The **database password** (`OutReachAgent1`) was used only inside a temporary script in the session's scratchpad folder, which has now been deleted. It is not stored in any project file. It is still visible in this chat's history, though, and in Supabase's own dashboard. If you'd like extra peace of mind, you can reset it from **Project Settings** → **Database** → **Reset database password** at any time — nothing in the codebase needs updating if you do, since the codebase itself never stored it.

8. What you should do right now

1 Look at your Supabase project

Table Editor (left sidebar) → you should see all 9 tables, with `accounts` holding 3 rows and `settings` holding 1. This is the same thing the smoke test just confirmed programmatically — seeing it yourself is worth 30 seconds.

2 Commit and push

Two small pieces of work happened this session: the `.gitignore` update from the config change, and this Phase 1 documentation. Nothing else needs committing — the three Phase 1 code files were already pushed last session under commit `60e8f23`.

```
git add .gitignore docs/PHASE_1_EXPLAINED.html docs/PHASE_1_EXPLAINED.pdf PROGRESS.md
REQUIREMENTS_COVERAGE.md
git commit -m "Phase 1 verified live – database schema and RLS running in Supabase"
git push
```

Note: `agent/.env` and `dashboard/.env` are correctly excluded — they were never staged and never will be.

3 Say the word for Phase 2

Phase 2 builds the account pool, isolated browser sessions per account and proxy, health detection, and the daily scheduler. This is one of the two phases in the build where the reasoning quality genuinely matters — a subtle mistake here risks a real LinkedIn or Instagram account getting flagged. Switch to Opus for it when we start (`/model opus`).

9. What's needed before Phase 2

Nothing new is required. Phase 2 builds the account pool manager and session isolation logic with proxy slots left empty — proxies aren't needed until Phase 10. You can start Phase 2 immediately.

Still open for later phases: a Claude API key (Phase 4) and a decision on the WhatsApp provider — Twilio, 360dialog, or the informal method (Phase 7).

Phase 1 of 11 complete and verified live · Generated 24 July 2026

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Tracking document: `REQUIREMENTS_COVERAGE.md` · Phase tracker: `PROGRESS.md`